Rank-27 Sweep Phase-Word Closure

Tactix Development Project

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Outcome

The general little-endian Sweep phase expansion now uses the eight masked 64-bit word values used by upstream Kitten, copied through alignment-safe `memcpy`. The scalar implementation remains the fallback on other byte orders. Exact tracing proves that random draws, all 64 phase bytes, rule and candidate order, ticks, scheduling, learning, pruning, and the resulting logical state are unchanged. Complete correctness, performance, proof, model, N1–N4, and Claude-owned fuzz gates pass. Rank 27 is therefore parked; the next convergence witness will come from a fresh current-Tactix/current-Kissat comparison over immutable ranks 1–40.

1 Implementation and boundary

`SubSolver::randomize_phases()` expands each random word into eight masked word values on little-endian targets and transfers the fixed 64-byte block with `memcpy`. This retains upstream's equal-work representation opportunity without relying on the alignment of `vector<uint8_t>`. The isolated solver diff based on predecessor `ec6ef1c` has SHA-256 `263cd847300c64764fe73d04cdcc78faf1b6b77ea1ca871458ddd65fd29e4463`. No benchmark identity, rank, deadline, policy arm, strategy constant, proof rule, or schedule enters solver behavior.

2 Exact common-prefix evidence

Frozen campaign `20260826-2320-rank27-sweep-phase-word-prefix`, array `1650366`, ran candidate and predecessor as separate tasks on node018 with 12 allocated logical CPUs, exactly 48,000 MiB scheduler memory, and a 16 GiB process cap. Both intentionally stopped at conflict 9,450 and reproduced normalized semantic SHA-256 `3b87bb36da711bcef3ca877f3922e9fa5f8a3c6ae9d638c1ae18d4965bd92c5f`.

The changed phase-expansion owner fell from 849,084,660 to 150,625,144 sampled cycles (82.26%); complete prefix cycles fell 1.1375%. Sweep time fell from 910 to 744 ms at the exact same 41,351 solves and 37,087,279 ticks. Perf lost no samples, and cleanup and certificate residue were zero.

3 Decision-grade Rank-27 timing

Campaign `20260826-2340-rank27-sweep-phase-word-timing`, array `1650368`, used three paired waves and six independent one-solver tasks on an otherwise empty node018. Every job used the same 12-CPU, 48,000 MiB, 16 GiB proof-off Release contract. All arms returned UNSAT with exact full normalized non-time SHA-256 `10372d94d2cb1ae86893408435e0c3ffa2338053b11b7012c2107a3bebf1ffd0`.

Wave	Candidate (s)	Predecessor (s)	Ratio
1	358.542	358.587	0.999875
2	357.916	358.759	0.997650
3	357.941	359.460	0.995774

The median paired ratio is 0.997650, a 0.2350% reduction, and the full ratio range is 0.995774–0.999875. Candidate and predecessor observations span only 0.1749% and 0.2433% of their medians; all candidate runs are below 400 seconds. Median cycles fall 0.2414% and instructions 0.4084%.

4 Complete B28+C5 guard

Frozen campaign `20260827-0010-BC-sweep-phase-word-guard` used a full-width, unthrottled, node-unconstrained 66-task array `1650774`; aggregate job `1650840` classified all rows. Each independent task ran one solver with 2 allocated logical CPUs, 16 GiB, and the 1,200-second contract. All 66 tasks passed: each arm independently checked 17 SAT models and 16 UNSAT proofs, with no invalidity, solve loss, malformed result, cleanup failure, or certificate residue. Candidate/predecessor PAR-2 is 1,362.562/1,361.275 seconds (1.000945x), within the 1.10 guard. The aggregate summary SHA-256 is `ac59145977a044af299d703c24ea5dc84bd25c25ded4ba85294891c5d6e4cae1`.

5 Claude-owned fuzz authority

Claude Code exclusively prepared, submitted, monitored, inspected, and classified `20260827-0040-sweep-phase-word-fuzz`; Codex did not inspect its fuzz artifacts. Claude's terminal verdict is PASS. Prep job `1651241`, generated legs `1651243`, `1651244`, and replacement random leg `1651247`, plus reach job `1651246`, used an immutable read-only harness snapshot. Each generated leg requested 50 physical cores (100 logical CPUs), 400,000 MiB, and 50 workers.

Claude reported 81/81 corpus replays and 30,000/30,000 generated cases with zero failure: 13,914 checked SAT models, 10,955 accepted native LRAT proofs, and current-Kissat agreement on all 24,869 verdict cases. The 5,131 bounded five-second nonterminations carry no oracle claim. Direct tracing reached the changed word expansion and unchanged scalar tail. There was no sanitizer, crash, invariant, model, proof, or differential failure, and no certificate or scratch residue.

Two general harness repairs were required by the immutable snapshot workflow: snapshot-executed `fuzz.py` now resolves checker paths through `TACTIX_REPO_ROOT`, and `slurm.sh` restores owner write permission before removing a successful read-only snapshot. They change no solver, input, seed, case, or oracle behavior; the replacement leg validates cleanup.

6 Documentation and next action

The living resume surface and engineering notes were compacted under the repository documentation policy. Unique experimental detail remains in this report, the frozen technical report, campaign manifests, aggregate summaries, and Git history. All 75 Release tests pass after compaction.

The end-to-end Rank-27 gain is deliberately small, but it closes the demonstrated equal-work owner safely. The accepted Tactix pin and accepted current-Kissat diagnostic comparator will next be frozen and run as independent one-solver jobs over immutable ranks 1–40 using 12 allocated logical CPUs and exactly 48,000 MiB per job. Every verdict will be independently validated. The largest meaningful UNSAT slowdown will be confirmed with three paired waves before its earliest portable common-state divergence is opened.